

Improving Depth Estimation for Robotic Disassembly

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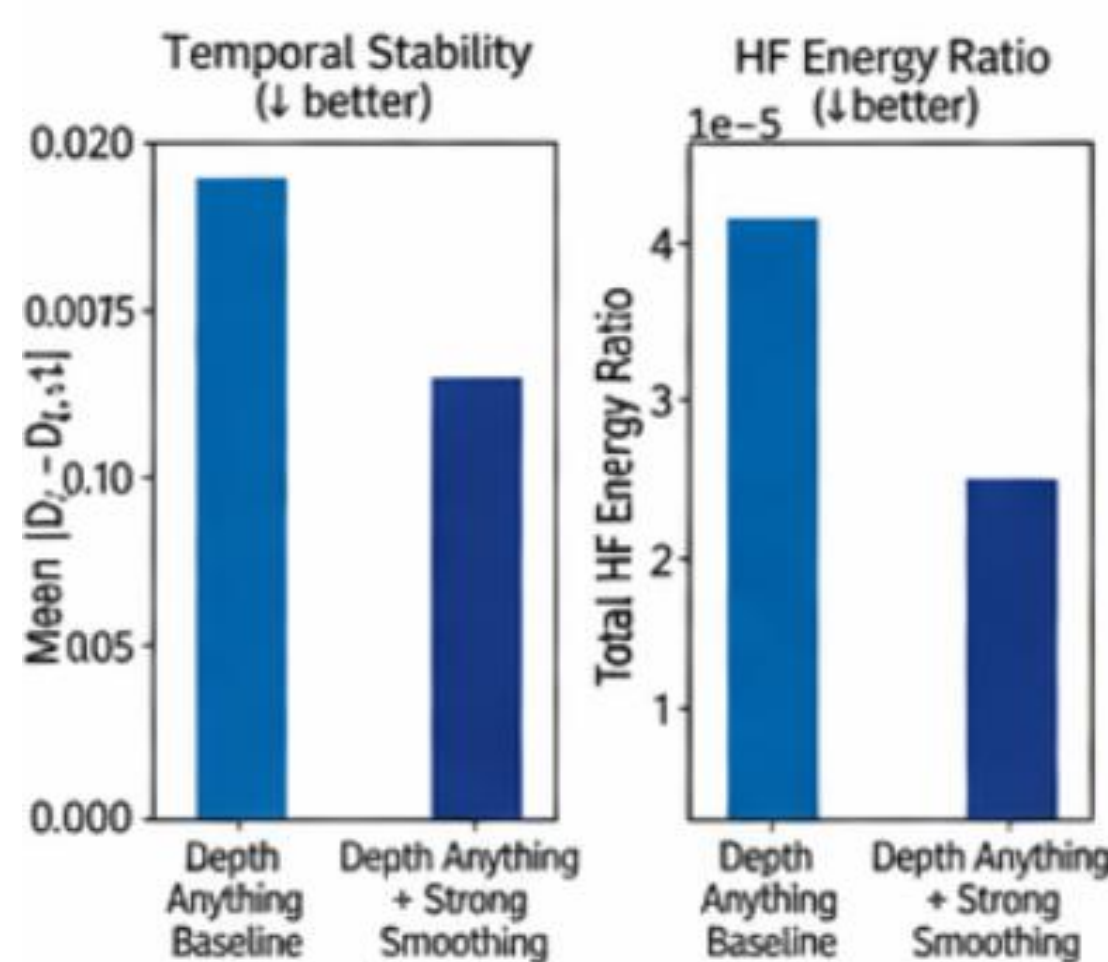
Motivation

- Depth estimation is critical for robotic disassembly tasks such as grasping and part separation
- Real-world scenes contain clutter, reflections, occlusions, and lighting variations
- Existing models perform well on benchmarks but fail in real environments



Evaluation Metrics

- Temporal Stability (Measures frame-to-frame consistency)
→ Lower is better
- High-Frequency Energy Ratio (Measures noise in depth maps)
→ Lower is better



Real-world Applications

- Robotic disassembly and Recycling automation
- Depth-based grasp planning and object manipulation
- Industrial inspection and part segmentation
- Safe human-robot interaction in cluttered environments

Method & Approach

Baseline Models

- MiDaS
- Depth Anything
- DepthCrafter

Selected: Depth Anything (Best Balance)

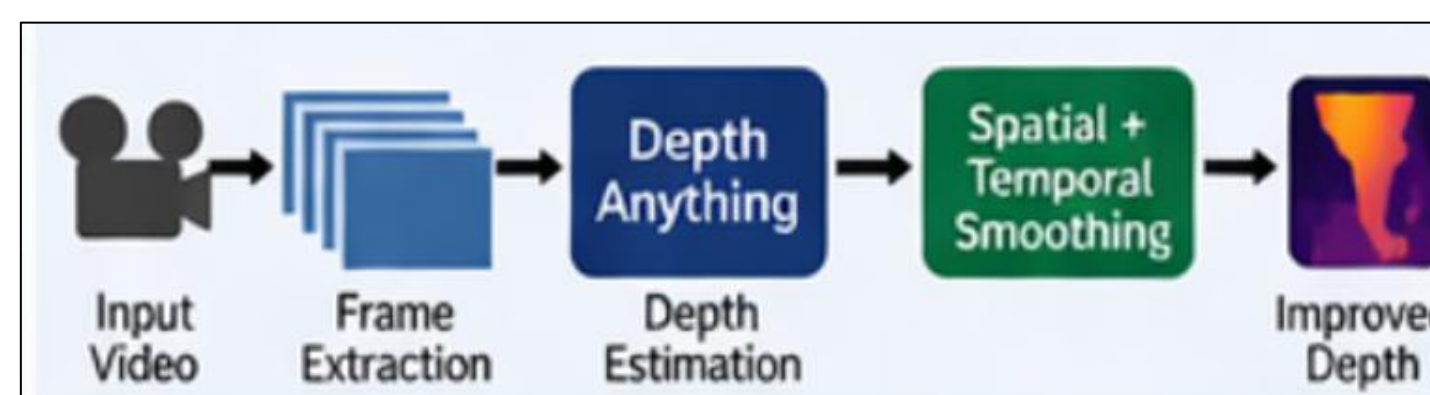
Attempted

- ✓ Fine tuning
- ✓ Self-supervised learning
- ✓ Teacher-based training
- ✗ Did not improve consistently

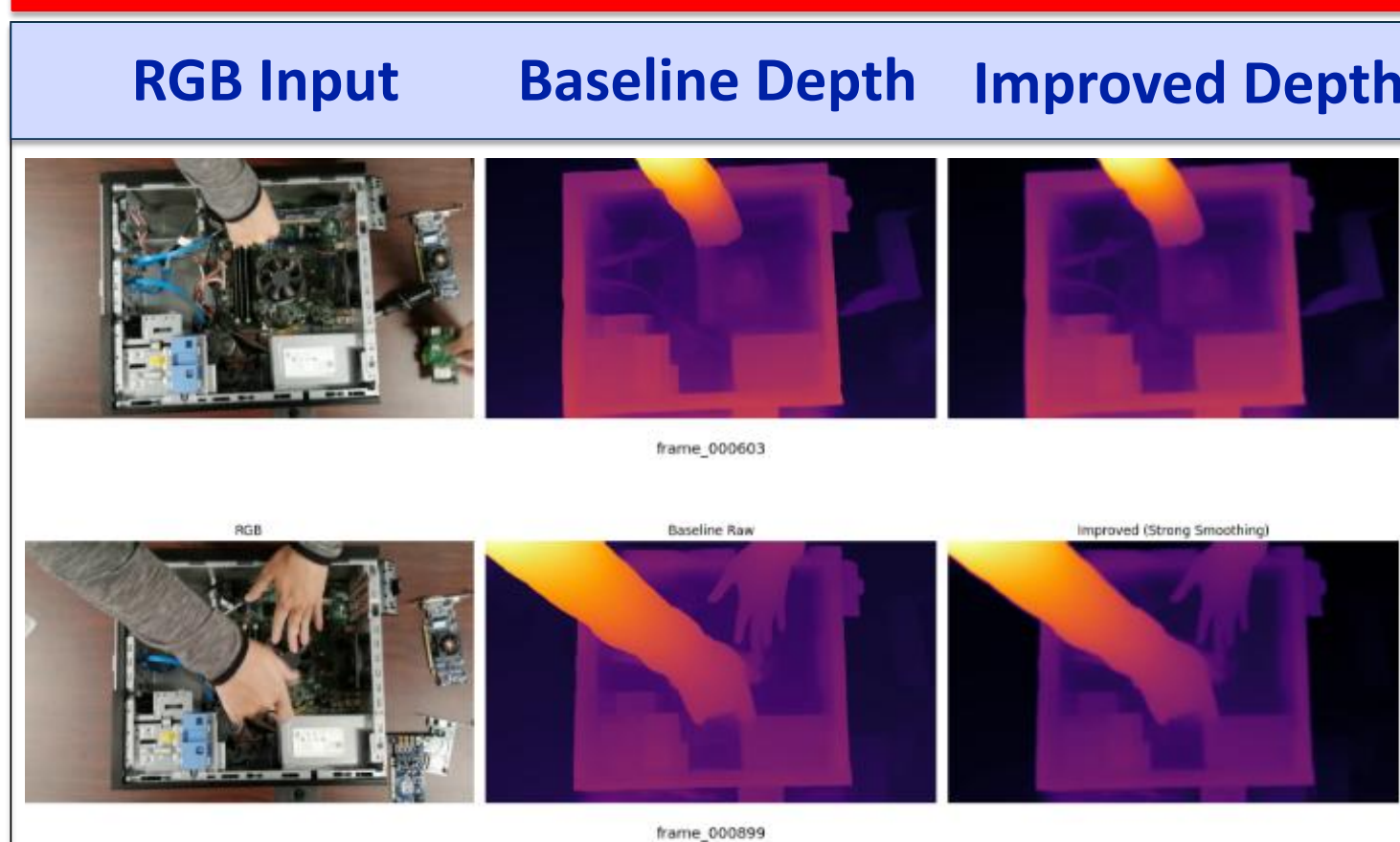
Final Solution

Strong Smoothing

- Gaussian spatial smoothing
- Temporal smoothing
- 16-bit depth precision



Final Solution: Strong Smoothing



Cleaner, smoother, and more stable depth maps

Smoothing Techniques Used:

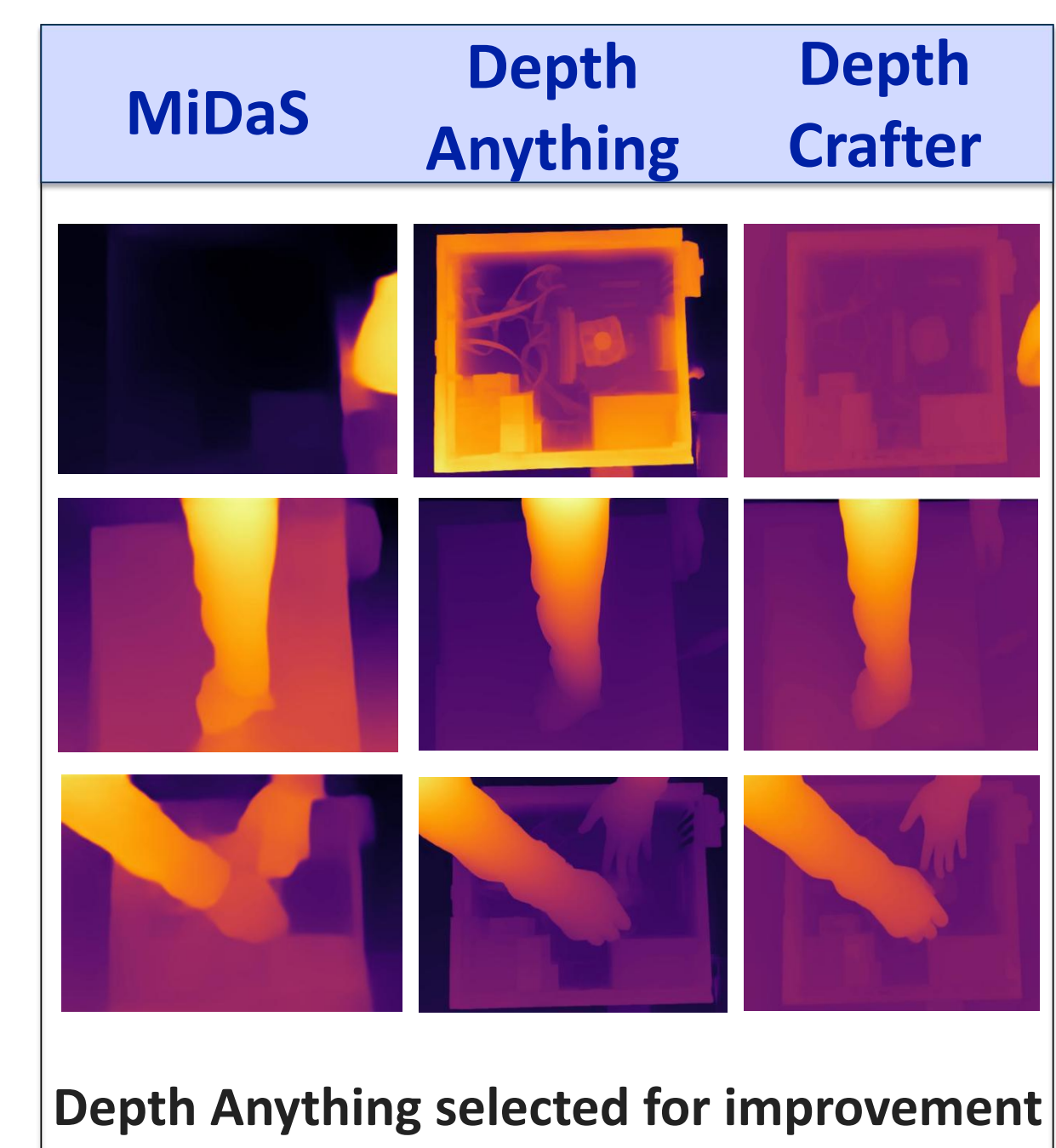
- Gaussian Spatial Smoothing – Reduces local noise
- Temporal Smoothing – Reduces flicker across frames
- 16-bit Depth Precision – Preserves fine details

References

- [1] Ranftl et al., "MiDaS: Robust Monocular Depth Estimation," 2022
<https://arxiv.org/abs/1907.01341>
- [2] Yang et al., "Depth Anything," 2024
<https://arxiv.org/abs/2401.10891>
- [3] Roboflow Blog – Depth Estimation Models
<https://blog.roboflow.com/depth-estimation-models/>

Baseline Comparision

- Depth Anything Showed the best balance of stability and structure
- MiDaS: tempral instability
- DepthCrafter: higher noise



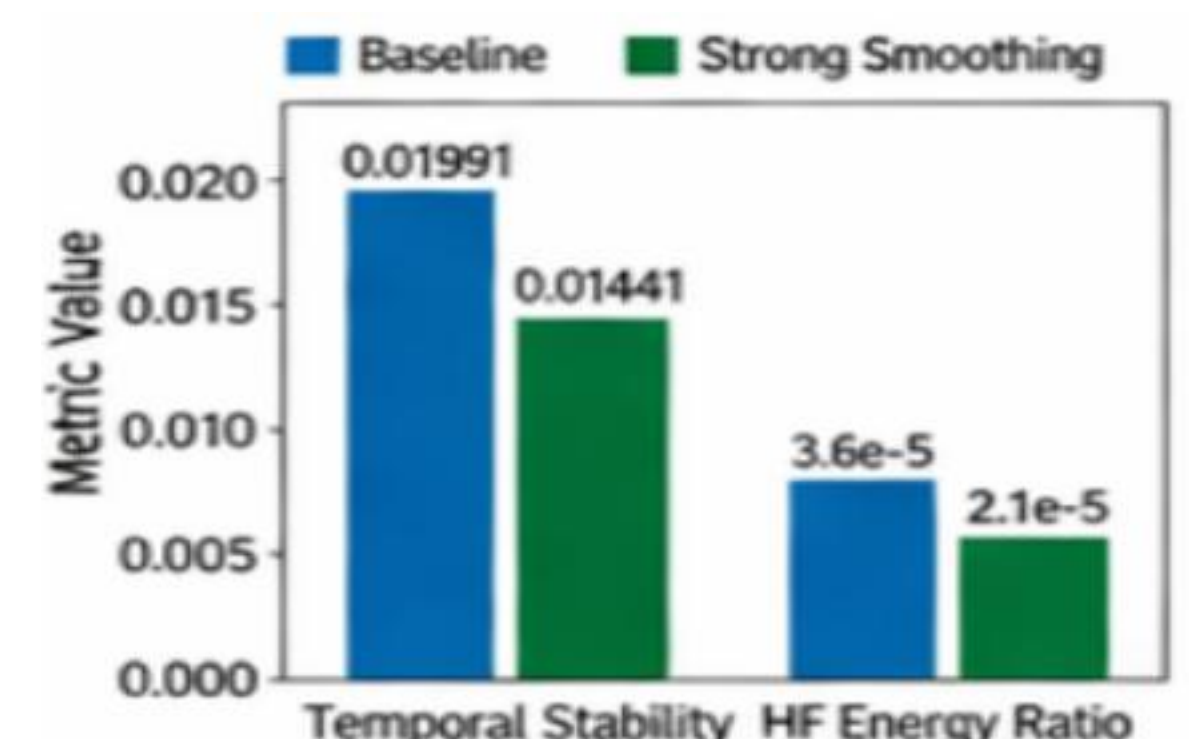
Depth Anything selected for improvement

Final results

Baseline Vs. Improved

- Temporal Stability: 0.01991 → 0.01441
↓ 27.6%
- HF Energy Ratio: 3.6e-5 → 2.1e-5
↓ 41.7%

- ✓ Reduced noise
- ✓ Improved Consistency
- ✓ More stable depth maps



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